

Assemble a Genome Project - *Ramazzottius varieornatus*

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Introduction

Within the Antman superhero movie trilogy, the concept of going subatomic, shrinking smaller than an atom to enter a quantum realm, is expressed firmly and plays a vital role in all three storylines. Specifically for the movie “Ant-Man and the Wasp”, directed by Peyton Reed, at one point a protagonist is halted midway in the processes of going subatomic due to a malfunction within the machinery. During that scene, multiple organisms of the same species called tardigrade are shown.

From the article, “Everything you need (and want) to know about tardigrades” by Laruen Robertson, tardigrades are described as “eight-legged segmented micro-animals” (Roberston, 2022) and were nicknamed the “little water bear” when first discovered by German zoologist Johann August Ephraim Goeze in 1773. The most intriguing fact is that tardigrades “have been subjected to all manner of supposedly unliveable conditions – from temperatures near absolute zero to crushing pressures and radiation that should easily kill them. They’ve even been thrown into the vacuum of space – all without batting an eyelid.” (Roberston, 2022). Given their indestructible nature, this organism was selected for genome assembly.

Methods

According to National Geographic, approximately 1,300 species of tardigrades have been discovered worldwide. For this project, the selected tardigrade species *Ramazzottius varieornatus* is a type of eukaryote with a genome size of 55.8 megabases and an unknown number of chromosomes.

The raw sequencing data utilized for this assembly was sourced from the Sequence Read Archive (SRA) under the experiment accession number DRX012456. The specific run utilized for the downstream assembly pipeline was DRR013910. This data was submitted by the National Institute of Genetics (Japan) under the study PRJDB1451. The overall goal of this sequencing project was to sequence the "Genome of anhydrobiotic extremotolerant tardigrade, ready to suspend," aiming to provide a comprehensive genomic foundation to understand the molecular mechanisms behind the organism's extreme survival capabilities, including anhydrobiosis (extreme desiccation tolerance).

The genome was sequenced using the Illumina Genome Analyzer IIx platform. This instrument utilizes sequencing by synthesis (SBS) technology, which uses fluorescently labeled reversible terminator nucleotides to determine the DNA sequence base by base as it is synthesized on a solid surface. The data output for the selected run (DRR013910) resulted in a 2 Gb file containing 20,531,102 short, paired end reads. In total, this run generated 3.1 billion bases to be used for the downstream de novo assembly.

Following data retrieval, the paired end reads (DRR013910) were imported into the Galaxy platform for quality assessment using FastQC, which revealed a degradation of base call quality at the 3' ends. To address this, Trimmomatic was utilized to perform a sliding window quality trim, removing low quality bases and ensuring only high fidelity data proceeded to the assembly phase, as verified by a secondary FastQC analysis. The cleaned reads were then processed through two distinct de novo De Bruijn graph assemblers for comparative analysis: SPAdes, selected for its strong performance with small eukaryotic genomes and built in error correction, and ABySS. The resulting scaffolds from both algorithms were evaluated structurally using

QUAST to compare metrics such as total length, largest contig size, and N50 score, while biological completeness was assessed against the eukaryota_odb10 lineage database using BUSCO. Finally, Bandage was used to visualize the SPAdes assembly graph, providing a graphical representation of contig connections and unresolved repetitive regions within the genome structure.

Results & Discussion

FastQC Analysis and Preprocessing

Initial quality assessment of the raw paired end sequence data was conducted using FastQC.

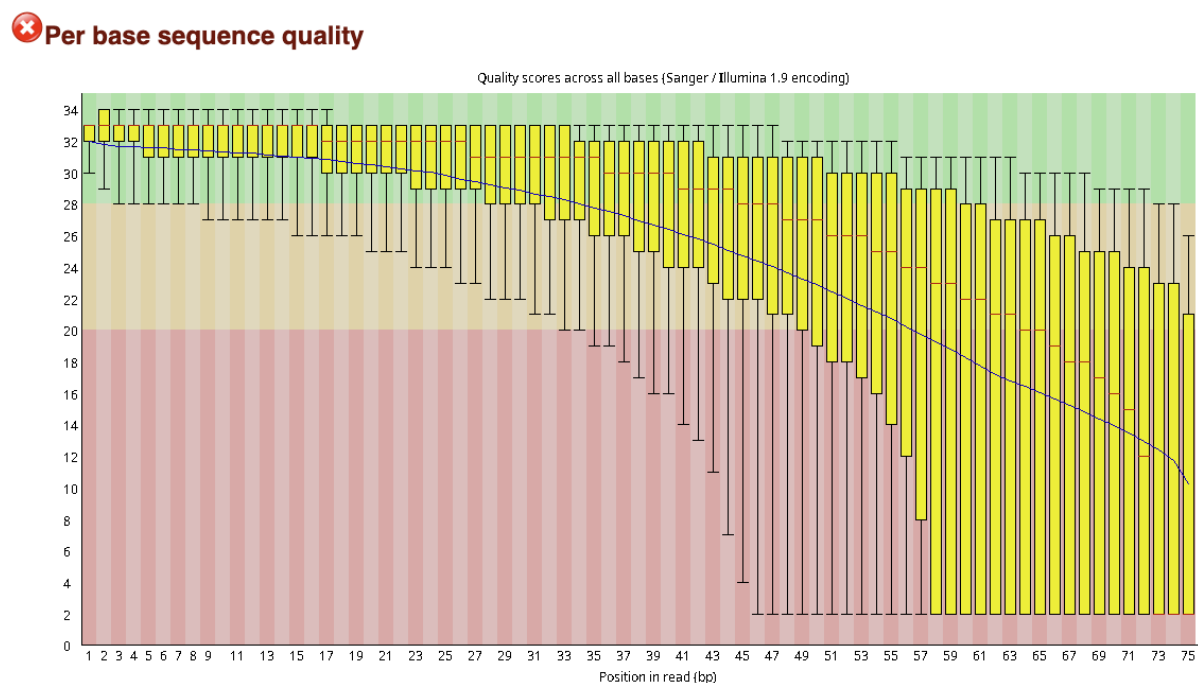


Figure 1. FAST QC Forward Strand Report Per Base Sequence Quality Failure

The initial FastQC report indicated that the majority of the sequence length maintained high Phred scores; however, the 'Per base sequence quality' module triggered a failure (Figure 1) because the sequence quality consistently degraded below a score of 20 at the extreme 3' ends. The following trend induced the same error for both forward and reverse strands.

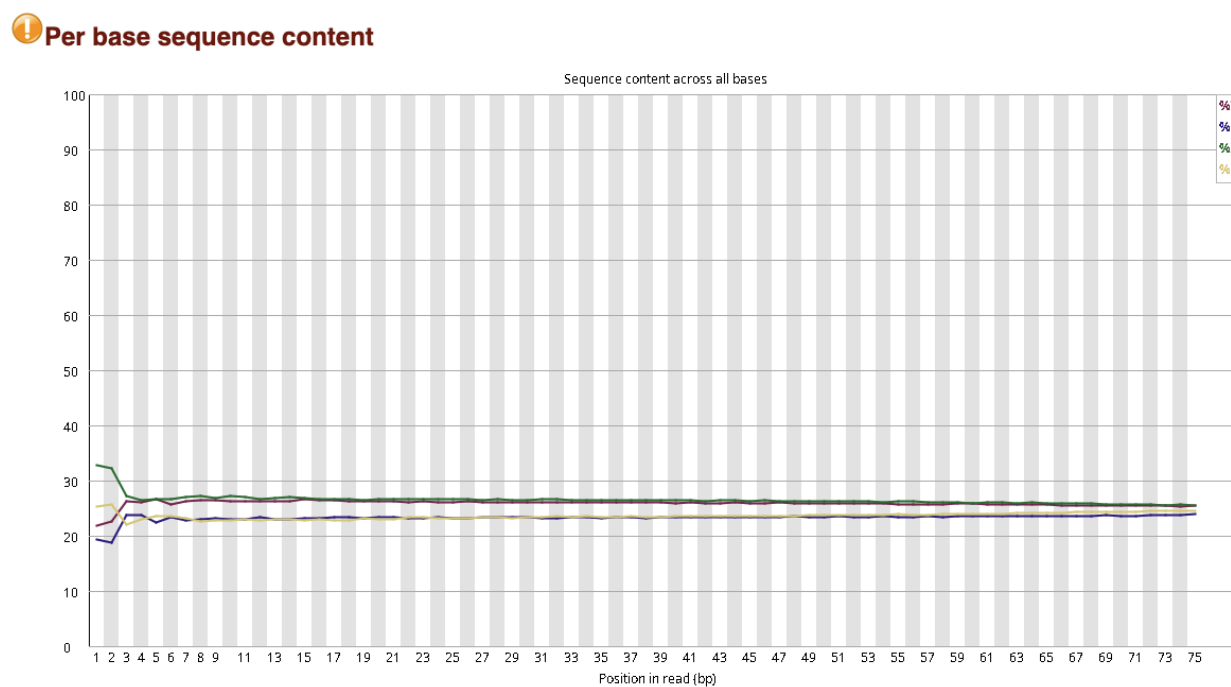


Figure 2. FAST QC Forward Strand Report Per Base Sequence Content Warning

The “Per base sequence content” warning (Figure 2) does appear for the forward strand but in comparison with the reverse, the percentage in which base is represented within the sequence remains consistent and is not unreasonable.

[illegible]

The initial FastQC report for the reverse strand flagged an overrepresented sequence consisting entirely of ambiguous 'N' bases (Figure 3), accounting ~0.16% of the reads. This is a common sequencing occurrence that represents uncalled bases. Since these bases are usually represented with bad quality scores, they are eventually removed once they are preprocessed. Due to the degradation of base call quality at the 3' ends could introduce errors into the sensitive De Bruijn graph assembly process, preprocessing was required. Trimmomatic was utilized to perform a sliding window quality trim and remove low quality bases, which includes the ambiguous bases. A secondary FastQC analysis was performed on the trimmed output, which confirmed the low quality trailing bases were successfully removed, resolving most of the previous warnings and ensuring a high fidelity dataset for assembly.

The only remaining flag post processing was a minor warning within the 'Per base sequence content' module for the forward strand. The graph representing this warning mirrors the original 'Per base sequence content' graph. Its persistence simply reflects the natural base composition bias of the original fragments and does not negatively impact the downstream assembly.

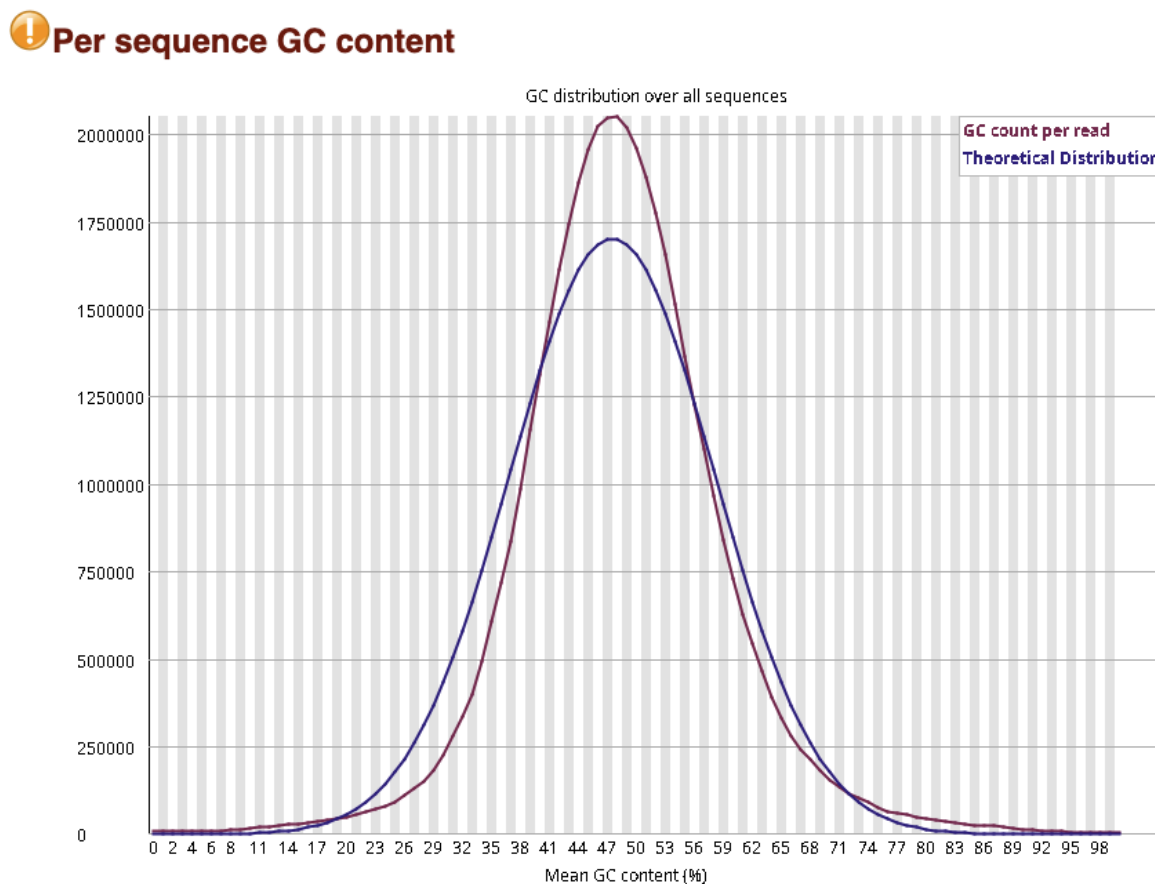


Figure 4. FAST QC Forward Strand Report Per Sequence GC Content

The two new warnings that appear for both forward and reverse strands are ‘Per sequence GC content’ and ‘Sequence length distribution’. Both warnings are expected of the trimming process and do not indicate poor data quality. The ‘Per sequence GC content’ warning (Figure 4) was triggered because the actual GC distribution of the reads was significantly narrower, displaying a lower standard deviation, than the theoretical normal distribution predicted by FastQC. This tight, unimodal peak indicates that the sequences are highly consistent and free of secondary contamination.

! Sequence Length Distribution

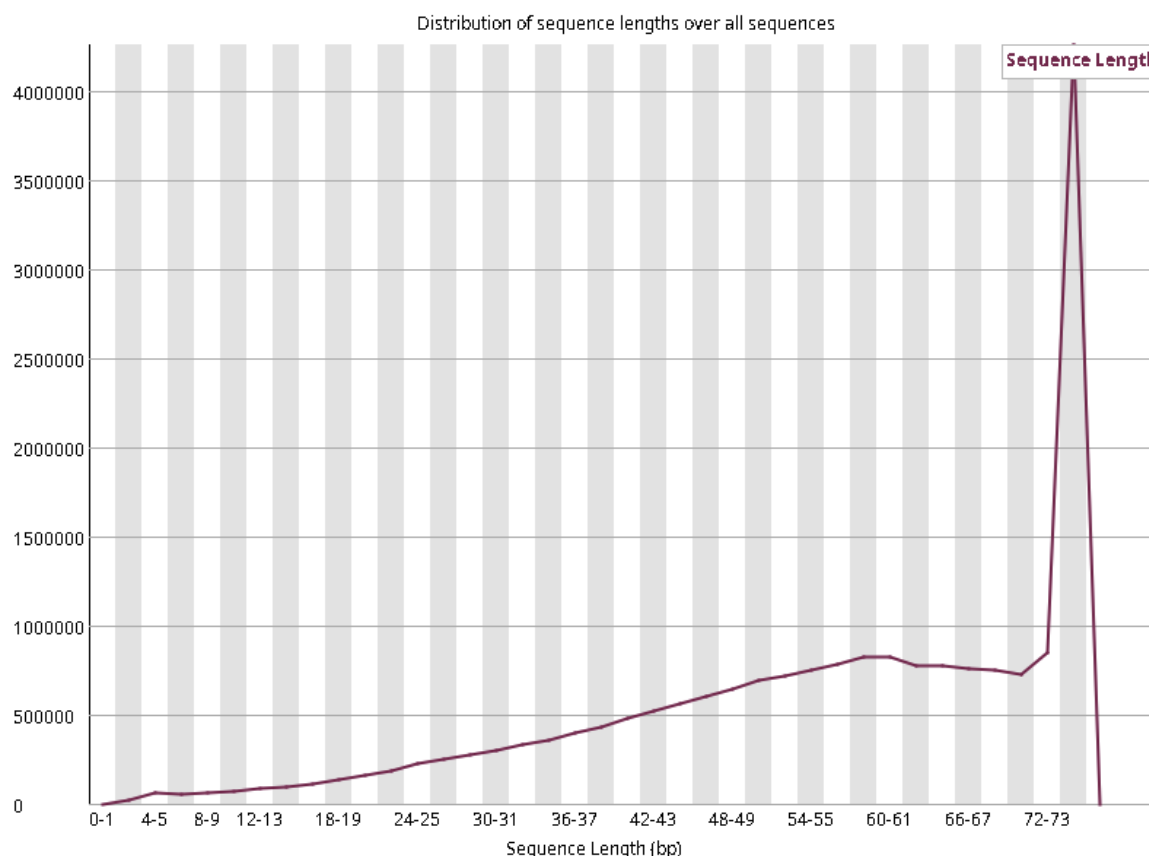


Figure 5. FAST QC Forward Strand Report Sequence Length Distribution

Additionally, the 'Sequence length distribution' warning (Figure 5) is a direct, intended result of the Trimmomatic tool. While the raw reads were a uniform 75 base pairs, the sliding window trim dynamically removed low quality trailing bases, resulting in variable read lengths across the dataset. FastQC flags this loss of uniformity by default, but it serves as confirmation that the low fidelity bases were successfully discarded.

De Novo Genome Assembly

The trimmed, high quality paired end reads were assembled using two distinct de novo algorithms: SPAdes and ABySS. Both programs were run using their default parameters optimized for short read paired data. The SPAdes algorithm successfully completed, generating the primary outputs: contigs (contigs.fasta), scaffolds (scaffolds.fasta), and an assembly graph (assembly_graph.fastg). Parallely, the ABySS algorithm generated its respective contigs, unitigs, and scaffolds files. Neither algorithm failed during execution.

Assembly Quality Assessment

To evaluate and compare the structural integrity of the assemblies, QUAST was utilized. Because a chromosome level reference genome is not currently available for *Ramazzottius varieornatus*, reference dependent metrics such as the 'percentage of reads mapped' could not be calculated. However, comparative structural statistics were successfully recorded:

- SPAdes Assembly: Generated a total of 6,422 contigs (≥ 500 bp), achieving a total assembly length of 51,997,756 bp. The largest assembled contig was 109,899 bp, and the assembly achieved an N50 of 16,626 bp.
- ABySS Assembly: Generated a total of 8,081 contigs (≥ 500 bp), with a total assembly length of 53,040,536 bp. The largest assembled contig was 110,433 bp, and the assembly achieved an N50 of 12,862 bp.

To contextualize these statistics, the N50 value, which indicates that half of the genome is contained in contigs of that length or longer, serves as a primary indicator of assembly contiguity. While both algorithms successfully reconstructed roughly 93% to 95% of the

expected ~55.8 Mb genome size, the SPAdes algorithm produced a noticeably superior structural assembly. SPAdes yielded significantly fewer individual fragments (6,422 vs. 8,081) and achieved a much higher N50 score, indicating a less fragmented and more continuous overall reconstruction.

Biological completeness was assessed using BUSCO against the eukaryota_odb12 lineage database.

- SPAdes BUSCO: The SPAdes assembly successfully completed the analysis, indicating a high degree of biological fidelity. Out of 129 conserved eukaryotic gene groups searched, the assembly recovered 86.0% as complete, single copy orthologs. Only 8.5% of the genes were fragmented, and 5.4% were missing entirely. Furthermore, the algorithm yielded a 0.0% duplication rate, indicating that it successfully avoided artificial sequence inflation during the graph resolution phase.
- ABySS BUSCO: The ABySS assembly failed to complete the BUSCO analysis. The job was forcefully terminated by the public galaxy server (Exit Code 137: Out of Memory). This computational failure suggests the highly fragmented nature of the ABySS scaffolds caused the hidden Markov models within BUSCO to exceed standard memory allocations during the gene mapping phase.

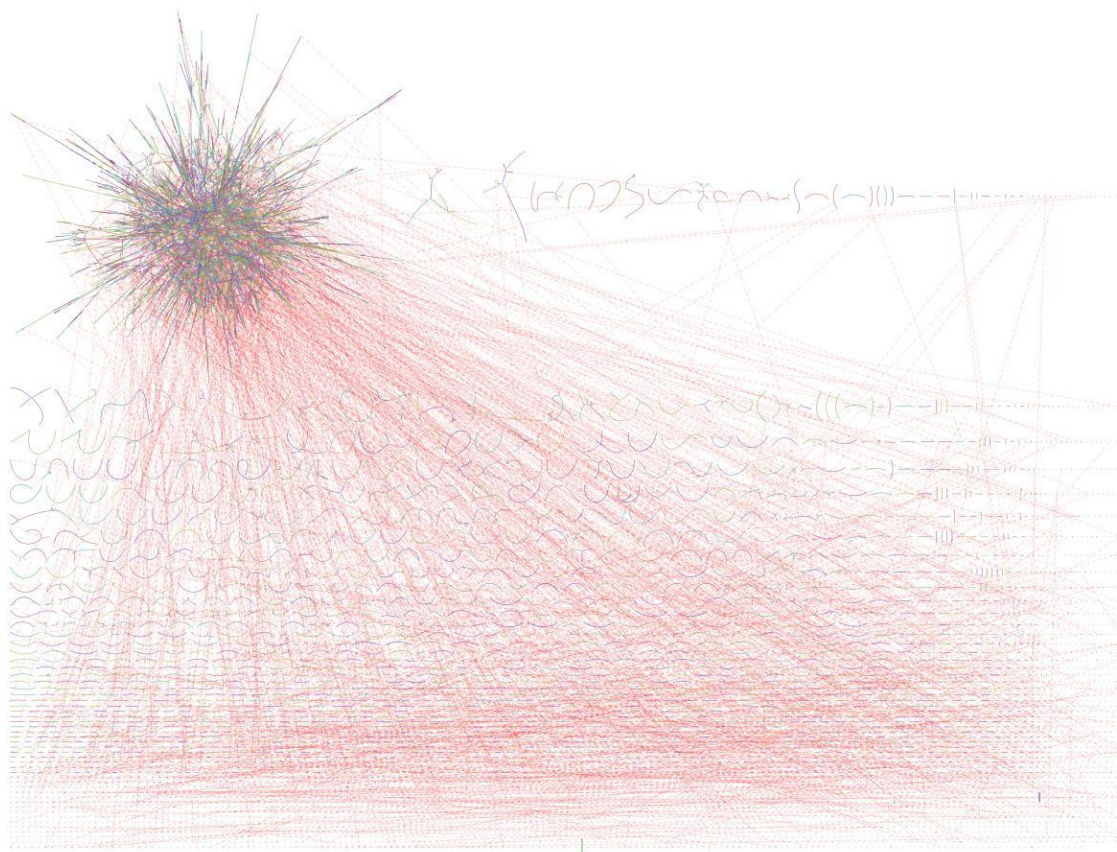


Figure 6. Bandage Visualization of the SPAdes de novo Assembly Graph

A visual assessment of the structure of the SPAdes assembly was conducted using Bandage (Figure 6). The generated assembly graph clearly illustrates the inherent challenges of short read de novo assembly. The visualization is dominated by a highly complex, dense central node cluster (top left). This structure represents unresolved repetitive genomic elements where the 75 bp short reads lacked the necessary length to unambiguously span the repeats, preventing complete graph resolution.

However, branching out from that messy center, you can see plenty of long, distinct, continuous paths. These lines represent the unique parts of the genome that assembled

successfully, and they visually explain why we got such a solid N50 score. Additionally, visible scaffolding linkages indicate where paired end reads successfully linked adjacent sequences, even without enough direct overlap for full contig merging.

Comparison of Assembly Algorithms

When evaluating both the structural and biological quality metrics, the SPAdes algorithm produced a vastly superior assembly compared to ABySS for this dataset. Structurally, SPAdes achieved a significantly higher N50 score (16,626 bp compared to ABySS's 12,862 bp) and produced a more contiguous overall reconstruction, yielding fewer individual fragments (6,422 contigs versus ABySS's 8,081).

This structural superiority was strongly confirmed during the biological completeness (BUSCO) analysis. The SPAdes assembly was contiguous enough to successfully map 86.0% of the core eukaryotic orthologs. In stark contrast, the ABySS assembly was so highly fragmented that it triggered a fatal Out of Memory (OOM) error on the server during the evaluation process.

Ultimately, SPAdes outperformed ABySS likely because of its use of multi sized De Bruijn graphs and its built in error correction. These features allow SPAdes to handle the localized sequences and repetitive regions typical of short read Illumina data far better than the standard approach used by ABySS.

Complete Workflow Architecture

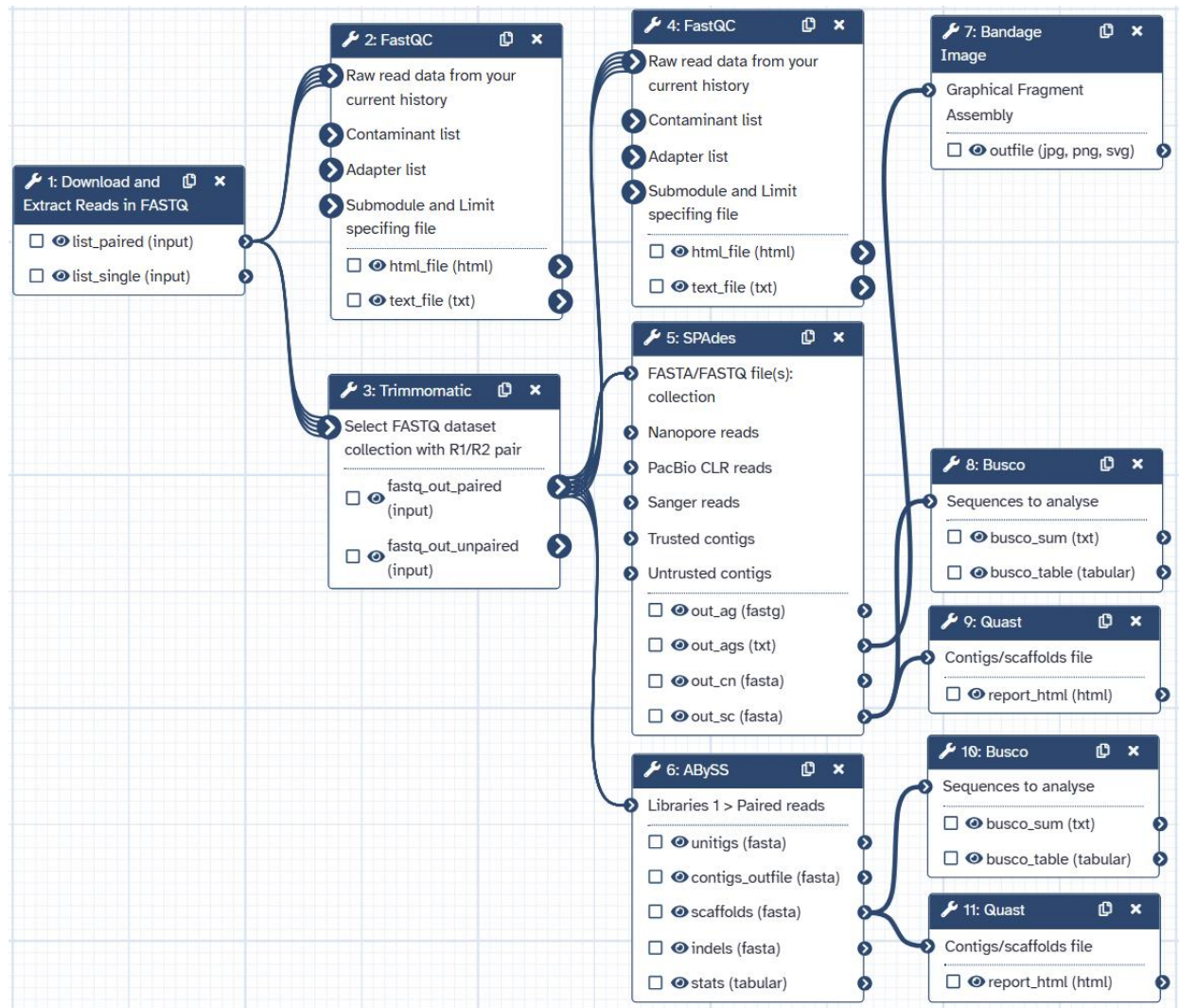


Figure 7. Workflow for the de novo Assembly of the *Ramazzottius varieornatus* Genome

The BUSCO biological completeness analysis was performed on the public Galaxy server (usegalaxy.org). All other computational steps in this pipeline, including read preprocessing, de novo assembly, and structural evaluation, were executed locally on the Biocomputational Engineering Program Lambda Vector GPU Systems instance of Galaxy.

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